

PROBLEM SET 5

PROBLEM - HEAT SINK

KNOWN: $r = .001\text{m}$ $L = .05\text{m}$ $s = .03\text{m}$ $L_1 = .02\text{m}$

$$T_\infty = 25^\circ\text{C} \quad T_b = 125^\circ\text{C}$$

$$h = 10\text{ W/m}^2\text{K} \quad k = 200\text{ W/mK} \quad k_1 = 10\text{ W/mK} \quad R_c = 10^{-4}\text{ m}^2\text{K/W}$$

FIND: (a) find $T(L/2)$, $T(L)$ (f) Thermal Resistance

(b) ϵ_f, η_f - Table 3.4 Network, $T_h, \Delta T$

(c) η_f - Table 3.5, discuss

(d) R_f

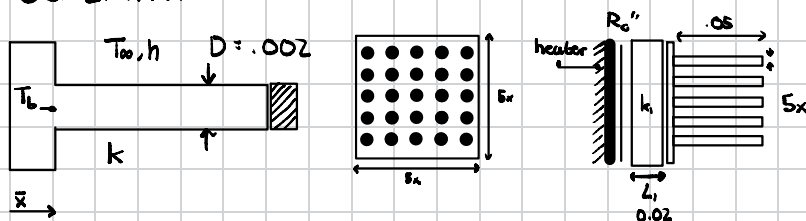
(e) $\eta_o, R_o, q_{o,f}, q_p$

ASSUMPTIONS: Steady state, 1D Heat Transfer, Constants

Negligible Thermal Radiation, Thermally Insulated,

No contact resistance

SCHEMATIC:



ANALYSIS:

(a) Using Fin Problem Workflow...

$$\text{Fin Equation: } \frac{\partial^2 T}{\partial x^2} - \frac{hP}{kA_c}(T - T_\infty) = 0 \quad \frac{d^2 \theta}{dx^2} - m^2 \theta = 0$$

$$\text{General Solution: } \theta(x) = T(x) - T_\infty = C_1 e^{mx} + C_2 e^{-mx}$$

$$\text{B.C: } T(0) = T_b \text{ (fixed temp)} \quad \theta(0) = \frac{e^{mx}}{e^{2mL} + 1} + \frac{e^{-mx}}{e^{2mL} + 1}$$

$$-k \frac{dT}{dx} \Big|_{x=L} = 0 \text{ (insulated tip)} \quad \theta(L) = \frac{e^{mx}}{e^{2mL} + 1} + \frac{e^{-mx}}{e^{2mL} + 1}$$

$$\begin{cases} C_1 + C_2 = \theta_b \\ \theta_b = T_b - T_\infty \end{cases}$$

$$m(C_1 e^{mL} - C_2 e^{-mL}) = 0$$

$$C_1 e^{mL} = C_2 e^{-mL}$$

$$C_1 e^{mL} = (\theta_b - C_1) e^{-mL}$$

$$C_1 (e^{mL} + e^{-mL}) = \theta_b e^{-mL}$$

$$C_1 (e^{2mL} + 1) = \theta_b$$

$$C_1 = \theta_b / (e^{2mL} + 1) \quad C_2 = \theta_b (1 - 1/(e^{2mL} + 1))$$

$$= \theta_b (e^{2mL} / (e^{2mL} + 1))$$

For $x = L$

$$= \frac{e^{mL} + e^{2mL - mL}}{e^{2mL} + 1}$$

$$= \frac{e^{mL} + e^{mL}}{e^{2mL} + 1}$$

$$= 2e^{mL} / (e^{2mL} + 1)$$

$$\frac{\theta(L/2)}{\theta_b} = .9146 \quad \frac{\theta(L)}{\theta_b} = .886$$

$$\theta(L/2) = 91.46\text{ K} \quad \theta(L) = 88.6\text{ K}$$

$$T(L/2) = 116.4^\circ\text{C} \quad T(L) = 113.6^\circ\text{C}$$

$$m^2 = hP/kA_c \text{ [1/m}^2\text{]}$$

$$P = 2\pi r, A_c = \pi r^2$$

$$m = 10\text{ m}^{-1}$$

$$\theta_b = T_b - T_\infty = 100\text{ K/}^\circ\text{C}$$

(b) From Table 3.4

Adiabatic

$$q_f = M \tanh mL \text{ where } M \equiv \sqrt{h P k A_c \theta_b} \text{ (as } k \rightarrow \infty$$

$$M = .628 \text{ W } A_f = PL + A_c$$

$$q_f = 0.29 \text{ W } = 3.1735 \cdot 4 \text{ m}^2$$

$$q_{max} = h A_f \theta_b = 0.317 \text{ W}$$

$$q_{nf} = h A_c \theta_b = .0031 \text{ W}$$

$$E_f = q_f / q_{nf} = 92.92$$

$$\eta_f = q_{max} / q_f = 0.915$$

(c) From Table 3.5

for rectangular pin-fin

$$\eta_f = \tanh mL_c / mL_c$$

$$\text{where } L_c = L + r/2$$

$$\eta_f = 0.923$$

This is within 0.85% of the η_f computed in part (b). This is a good approximation

$$(d) R_f = \frac{1}{\eta_f h A_f}$$

$$= 344.93 \text{ K/W}$$

$$(e) A_b = s^2, A_{uf} = s^2 - 25 A_c$$

$$A_{tot} = N A_f + A_{uf}$$

$$q_{tot} = N \eta_f A_f h \theta_b + h A_{uf} \theta_b$$

$$q_{tot, max} = N A_f h \theta_b + h A_{uf} \theta_b$$

$$\eta_o = \frac{q_{tot, max}}{q_{tot}} = \frac{8.754 \text{ W}}{8.08 \text{ W}}$$

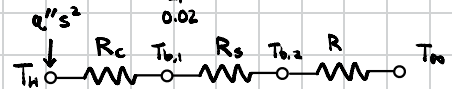
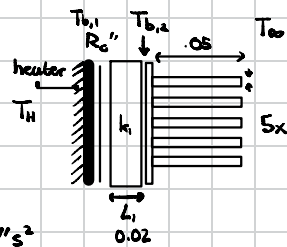
$$\eta_o = 0.9223$$

$$R_o = \frac{R_f}{25 R_{eff}} = \left(\frac{25}{R_f} \right) = R_f / 25 = 13.78 \text{ K/W}$$

$$R_{conv} = \frac{1}{h A_{uf}} = 121.73 \text{ K/W}$$

$$R_o = (13.78 + 121.73)^{-1} = 12.37 \text{ K/W}$$

(f) Thermal Network



$$(T_H - T_{\infty}) = q'' s^2 \cdot (R_c + R_o + R_s)$$

$$T_H = q'' s^2 \cdot (R_c + R_o + R_s) + T_{\infty}$$

$$\text{Scenario 1 } R = R_o \text{ (e)} = 12.37 \text{ K/W}$$

$$\text{Scenario 2 } R = \frac{1}{h s^2} = 111.11 \text{ K/W}$$

$$q'' s^2 = 0.9 \text{ W}$$

$$R_c = (10^{-4} \text{ m}^2 \text{ K/W}) / (9 \cdot 4 \text{ m}^2) = .11 \text{ K/W}$$

$$R_o = \frac{L_1}{k s^2} = 5.56 \text{ K/W}$$

$$T_{H,1} = 41.2^\circ \text{C (w/ heatsink)}$$

$$T_{H,2} = 130.1^\circ \text{C (w/o heatsink)}$$

$$q_f = \theta_b / 13.78 = 7.26 \text{ W}$$

$$q_{uf} = \theta_b / 123.73 = 0.82 \text{ W}$$

PROBLEM 2 - WIRE

KNOWN: $D = .001 \text{ m}$, $T_{\infty} = 20^{\circ}\text{C}$ $R_e = .01 \Omega/\text{m}$

$\rho = 4000 \text{ kg/m}^3$ $k = 400 \text{ W/mK}$ $c = 500 \text{ J/kgK}$

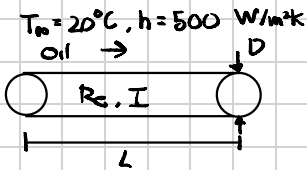
$h = 500 \text{ W/m}^2\text{K}$

@ $t = 0$, $I = 100 \text{ A}$

FIND: (a) T_{ss} (b) $T(t)$, T_{sc}

ASSUMPTIONS: Steady state, Negligible Radiation, Uniform temperature -
(1) lumped capacitance. Wire starts with $\theta_i = 0$ (2) (3)

SCHEMATIC:



ANALYSIS:

$$(a) \dot{E}_{st} = \dot{E}_{gen} - \dot{E}_{out} \text{ [W]}$$

$$\rho V c \frac{dT}{dt} = I^2 R_e - h A_s (T - T_{\infty})$$

$$\frac{\rho V c d\theta}{dt} = I^2 R_e - h A_s \theta$$

$$\rho V c \dot{\theta} + h A_s \theta = I^2 (R_e L)$$

$$\frac{\rho V c}{h A_s} \dot{\theta} + \theta = \frac{I^2 R_e L}{h A_s}$$

$$\theta_{ss} = \frac{I^2 R_e L}{h A_s} \text{ [K]}$$

$$= \frac{I^2 R_e L}{h L D \pi}$$

$$= \frac{I^2 R_e}{h D \pi}$$

$$= 63.66 \text{ K}$$

$$T_{ss} - T_{\infty} = \hat{\theta}$$

$$T_{ss} = 83.66^{\circ}\text{C}$$

$$\theta(t) = \theta_{ss} (1 - e^{-t/\tau})$$

$$= \theta_{ss} (1 - \exp(-\frac{h A_s}{\rho V c} t))$$

$$\frac{A_s}{V} = \frac{4 \cdot D \pi \cdot \frac{L}{D}}{\pi D^2 \cdot \frac{L}{D}} = \frac{4}{D}$$

$$\theta(t) = \theta_{ss} (1 - \exp(-\frac{4h}{\rho D c} t))$$

$$T(t) = 63.66 (1 - e^{-t}) + 20 \text{ [}^{\circ}\text{C]}$$

$$\theta_{ss} - 5 = \theta_{ss} (1 - e^{-t_5})$$

$$-5 = -\theta_{ss} e^{-t_5}$$

$$\frac{5}{\theta_{ss}} = e^{-t_5}$$

$$t_5 = \ln \left(\frac{63.66}{5} \right)$$

$$= 2.54 \text{ s}$$